

TOW-IDS: Intrusion Detection System Based on Three Overlapped Wavelets for Automotive Ethernet

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Abstract—Devices that ensure vehicle and driver safety or provide services to drivers generate a substantial amount of network traffic. The traffic is transmitted to the In-Vehicle Network (IVN) depending on the defined function. Consequently, to quickly process a lot of traffic transmitted to the IVN, an advanced network protocol such as Automotive Ethernet is necessary. However, owing to the connectivity reinforcement between devices inside a vehicle and external networks, attack vectors and vulnerabilities can be easily inherited from an established Ethernet to Automotive Ethernet. The present study proposes a method for detecting and identifying abnormalities in Automotive Ethernet based on wavelet transform and deep convolutional neural network. First, we define attack scenarios and extract normal and abnormal data corresponding to these scenarios. Second, we conduct several preprocesses, such as fixing the packet size and normalizing the network image data. Finally, we conduct extensive evaluations of the proposed method's performance, considering the size of network image data and multi-resolution levels. The results demonstrate that the proposed method can effectively detect an abnormality. Furthermore, the results suggest that our method is more effective in terms of time-cost compared to default ResNet and EfficientNet methods.

Index Terms—Intrusion detection system, in-vehicle network, automotive ethernet, deep learning, wavelet transform, data reduction.

I. INTRODUCTION

IN RECENT years, the automotive industry has seen rapid adoption of in-vehicle systems (such as infotainment system and advanced driver assistance system (ADAS)) and a multitude of sensors (such as a vision camera, radar, and lidar) that can accurately and quickly recognize the area around a vehicle. These advanced technologies generate a large amount of network traffic during communication between networks inside the system and communication with networks

outside the vehicle. As a type of integrated sensing and communication (ISAC) concept, a new network environment is required to enable fast and accurate integrated processing of data of different types of protocols generated from heterogeneous networks of a vehicle [1]. Currently, many studies and industrial efforts toward handling a large amount of network traffic in an Automotive Ethernet are underway. Furthermore, to secure compatibility with existing web or mobile services, Automotive Ethernet adopts current ICT communication protocols, including the original Transmission Control Protocol (TCP)/Internet Protocol (IP). However, this indicates that various attack vectors and vulnerabilities that had occurred in the existing communication environment can be conveyed in the Automotive Ethernet environment. Even a minor risk posed by the attack vectors and vulnerabilities can jeopardize the safety of drivers, passengers, and pedestrians around vehicles at any time. Therefore, to ensure security of the IVN and the safety of drivers, a study needs to be conducted to detect the abnormality of a vehicle network and to identify their attack types in the Automotive Ethernet environment.

These types of studies, generally referred to as anomaly detection and attack identification of the IVN, are being conducted based on message traffic, which is the data link layer of the Controller Area Network (CAN) protocol. The CAN bus is vulnerable to security issues that infringe confidentiality, integrity, and availability because it is more focused on handling errors. These issues have been proven by the various types of vulnerabilities and attack cases that have been occurring in the vehicle network. Moreover, as mentioned above, in a system environment such as a connected car, or autonomous vehicle in the near future, a large amount of network traffic will be generated because of the many sensors and systems embedded in the vehicle [2]. However, it is difficult to process large amounts of network traffic using the CAN protocol, which is the method of communication between electronic devices in a vehicle, because of its low handling performance. To surmount these problems, Automotive Ethernet, which includes gigabit Ethernet capability, is becoming the core element of IVNs as it can handle enormous amounts of network traffic with relative ease [3].

Concretely, our main contributions can be summarized as follows:

- 1) We propose a method for detecting and identifying abnormalities in Automotive Ethernet based on wavelet transform and customized DCNN, which shows a high

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performance in image classification. This framework focuses on the data created between heterogeneous networks in the Automotive Ethernet network based on actual equipment that can be installed in the vehicle.

- 2) Based on three protocols (*i.e.*, Audio-Video Transport Protocol (AVTP), generalized Precision Time Protocol (gPTP), and CAN protocol), we have efficiently designed the proposed algorithm. To support its evaluation, we defined the five attack scenarios capable of immediately impairing the functions of the IVN and sufficiently interfering with driving.
- 3) We conducted extensive evaluations of the proposed method's performance by considering different attack scenarios, data size, and multi-resolution levels through several preprocess such as fixing the length of the packet size and normalizing the network image data.

The rest of this paper is organized as follows. Section II reviews the preliminaries and related studies. Section III provides five attack scenarios based on three protocols. Section IV presents our method of intrusion detection based on wavelet transform. Section V describes a detailed implementation of our proposed approach and its evaluation. Finally, Section VI provides the concluding remarks.

II. PRELIMINARIES AND RELATED WORK

A. Automotive Ethernet

Owing to the introduction of various advanced technologies in vehicles, the need for Ethernet as an IVN configuration has been growing increasingly. Existing IVNs such as the CAN and Local Interconnect Network (LIN) protocols have various advantages such as cost effectiveness; reliability owing to long-term use; and adequate performance for many applications. However, Automotive Ethernet's introduction as an IVN configuration does not indicate that it completely replaces existing IVNs. The aim of Automotive Ethernet is to unify the in-vehicle communication network technology, which is currently a mix of different types of communication network technologies, into Ethernet technology. Also, the introduction of Automotive Ethernet does not aim to replace IVNs, which have previously had no problems with the Ethernet technology. Instead, it aims to adequately supplement existing technologies in areas such as a large bandwidth requirement [4], [5].

Automotive Ethernet consists of the following main protocols: AVTP, gPTP, and CAN protocol. AVTP is a protocol designed to transport audio, video, and control data to be transported over a Time-Sensitive Networking (TSN) capable network. It consists of the audio-video bridging (AVB) talker (*i.e.*, a source or producer of an AVTP stream) and AVB listener (*i.e.*, a destination, receiver, or consumer of an AVTP stream). AVTP is synchronized with one or more linked AVB listeners for transmitting streams from the AVB talker depending on gPTP [6], [7]. Second, PTP is a network-based time synchronization communication protocol used to synchronize devices (*e.g.*, audio and video) with sub-microsecond (μs) time accuracy [8]. PTP is divided into PTPv1 and PTPv2. PTPv2-based gPTP is modified for vehicle-use to make the network timestamp the same between various devices installed

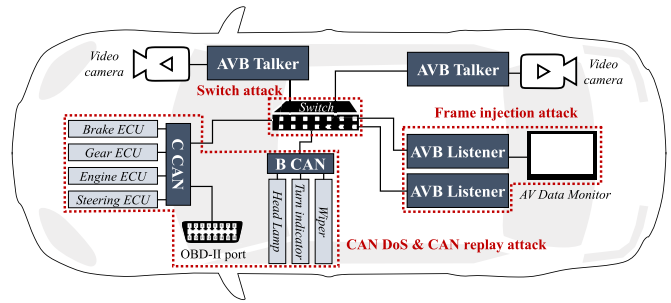


Fig. 1. Attack vectors in automotive ethernet environment.

in a vehicle in Automotive Ethernet. gPTP performs precision time synchronization of all endpoints included in the AVB network and provides the time information required for audio or video stream transmission. Namely, when looking into the transmission process of the AVTP stream, the AVB talker and listener, via gPTP, synchronize the time delay of propagation between a sender and receiver [9], [10]. Third, the CAN protocol is used in Automotive Ethernet networks as the core method communication between the various devices of a vehicle. This protocol transmits and receives CAN messages through ECU nodes parallel in the CAN bus. In CAN protocol, all ECU nodes connected to the CAN bus communicate with each another by acting as a master without a central master unit [11]. Further, there is no address information about the sender and receiver in the frame of the CAN message transmitted from an ECU node. If the transmitted CAN message relates to the receiving an ECU node, the message is accepted; otherwise, the message is ignored [12]. When several sending ECU nodes transmit a CAN message to a specific receiving ECU node simultaneously, the transmitted messages are determined based on the priority of the CAN messages by calculating their arbitration ID values by bit [13], [14].

B. Intrusion Detection of Heterogeneous In-Vehicle Networks

Studies from different perspectives on security systems and solutions are being conducted to reinforce vehicle security and protect drivers and passengers from vulnerabilities. Among vehicle IDS systems, specifically, the IDS research and implementation in the CAN bus network system is the most in-depth and long-studied field. Since most ECUs in the CAN bus system act as the core power unit of the vehicle, existing IDS studies can be said to be the groundwork for the security of various newly added protocols for supporting lots of sensors [15]. Currently, academic research on vehicle IDS mainly focuses on the taxonomy of IDS for CAN bus networks among protocols applied to vehicles. It is defined and explained as four taxonomy of deployment strategies, attacking techniques, technical challenges, and detection approaches [16]. The core of the main technology is the CAN message. The technology is divided into Frequency-based IDS, Content-based IDS, Combined-based and so forth. Among them, the frequency-based IDS is one of the typical methods for detecting anomalies by measuring the sequence of N consecutive CAN IDs output on the CAN bus based on the CAN message [17], [18]. Some research institutes

and companies are trying to establish standards, such as the Quality of Service (QoS) on the Automotive Ethernet, to satisfy functional requirements [19]. Studies on encryption and authentication technologies, such as Transport Layer Security (TLS) and Internet Protocol Security (IPSec), to ensure the integrity and confidentiality of communication messages in the Automotive Ethernet have also been proposed [20]. In addition, studies on the separation of the major domains of a vehicle and controlling access to a vehicle by using Virtual LAN (VLAN) are also being conducted [21], [22]. Unlike industrial studies, academic research results on Automotive Ethernet security show a gradually increasing tendency these days. In the recently published Automotive Ethernet study, a single protocol for the AVTP based on simulation data was studied [23]. In general, domain knowledge between heterogeneous networks was often required to perform intrusion detection via existing machine learning algorithms. In particular, if the topology or traffic pattern of the test data is significantly different from the training data, the algorithm may not perform consistently. To overcome these problems, the discussion of proposing models by applying various deep learning methodologies to raw data is continuously increasing [24]. The recently proposed Graph Neural Network (GNN) model by [25] exploits the relationships among flows, links, and queues in switched Ethernet networks and generalizes invisible topologies and traffic patterns for ameliorating the robustness of predictions. However, while the structure of the GNN model and the learning process is being performed, the time-cost of the prediction execution for an invisible configuration may increase restrictively.

Mai and Navet suggested that it is possible to overcome shortcomings with the optimization methods such as batch normalization or drop-out in a deep learning model [26]. Furthermore, In the case of the IDS for heterogeneous networks, like Automotive Ethernet, AutoEncoder is used to detect anomalies from high-dimensional sensor data of various data types, including image/video and sequence and graph data. In the encoding process, the encoder plays the leading role in mapping the original data into the low-dimensional functional space. On the other hand, the decoder function as the prime position for retrieving the original data from the projected low-dimensional space [9]. In other words, AutoEncoder has the advantage of the learning process, which takes into only high-priority data features from input data. In contrast, our framework focuses on the integrated data created between the heterogeneous networks, including AVTP, gPTP, and UDP which are converted from CAN messages, considering various sensing and communication devices that can be installed inside and outside a vehicle.

III. ATTACK SCENARIOS

This section covers five attack scenarios that can deepen the intensity of an attack and the degree of damage to Automotive Ethernet. As shown in Figure 1, the attack scenarios are based on the attack vectors that can occur in the Automotive Ethernet environment. These include: frame injection, PTP sync, switch (MAC flooding), CAN DoS, and CAN replay attacks. Figure 2

illustrates the five types of attack scenarios reproduced in detail in concordance with three main protocols, *i.e.*, AVTP, gPTP, and CAN. Detailed descriptions of the attack scenarios are provided below.

A. Frame Injection Attack

Frame injection attack refers to a form of attack that continuously injects back into the normal output stream after extracting an MPEG frame corresponding to a specific part of a typical video stream. The hex data in the header of the MPEG frame starts with '47 40', and this part can be extracted and inserted as video content for the attack into the middle of the MPEG frame. The stopped state image is distorted and displayed on the monitor connected to the AVB listener owing to the stream injected by the attack (see Figure 2a). If this attack occurs in an autonomous driving environment where a driver's cognition is not required, the vehicle will mechanically malfunction because the detection system of the IVN cannot identify and detect external objects around the driving vehicle [23].

B. PTP Sync Attack

To provide audio or video (*i.e.*, time-sensitive services) services in real-time without loss of information, generally, it must simultaneously synchronize all network devices on the data transmission path. When audio and video data are transferred, latency may occur in the switch, which is the primary component of an Ethernet network, owing to buffering. Time synchronization between vehicle network devices can be performed through PTP of the IEEE 1588 standard [27]. This standard provides a protocol that enables networked devices to utilize the most precise and accurate clock synchronization. The PTP sync attack refers to an attack that causes a time delay in the time synchronization process by modulating or flooding the time information of the sync message during the PTP synchronization process (see Figure 2b). This attack makes the time synchronization between master-slaves (*i.e.*, AVB talker and listener) impossible after injecting an incorrect sync message during the initial synchronization of the PTP on the Automotive Ethernet network [28].

C. Switch Attack (MAC Flooding)

A switch is to route network packets between heterogeneous networks. Similarly, it is also required for routing various network packets in Automotive Ethernet. In general, a switch is a device that follows a fail-open-policy that allows specific elements in the event of a traffic, permission, or service failure. If there is a problem in the system itself, the switch can operate as a type of hub. Consequently, if an attacker sends a large amount of frame packets to the switch, all the Media Access Control (MAC) addresses coming into the switch will be stored in the buffer of the MAC table. Eventually, it leads to overflow and the switch functions as a hub. Alternatively, an environment is created in which other hosts can sniff all the frame packets being broadcasted [29]. Owing to this attack, the Automotive Ethernet network becomes

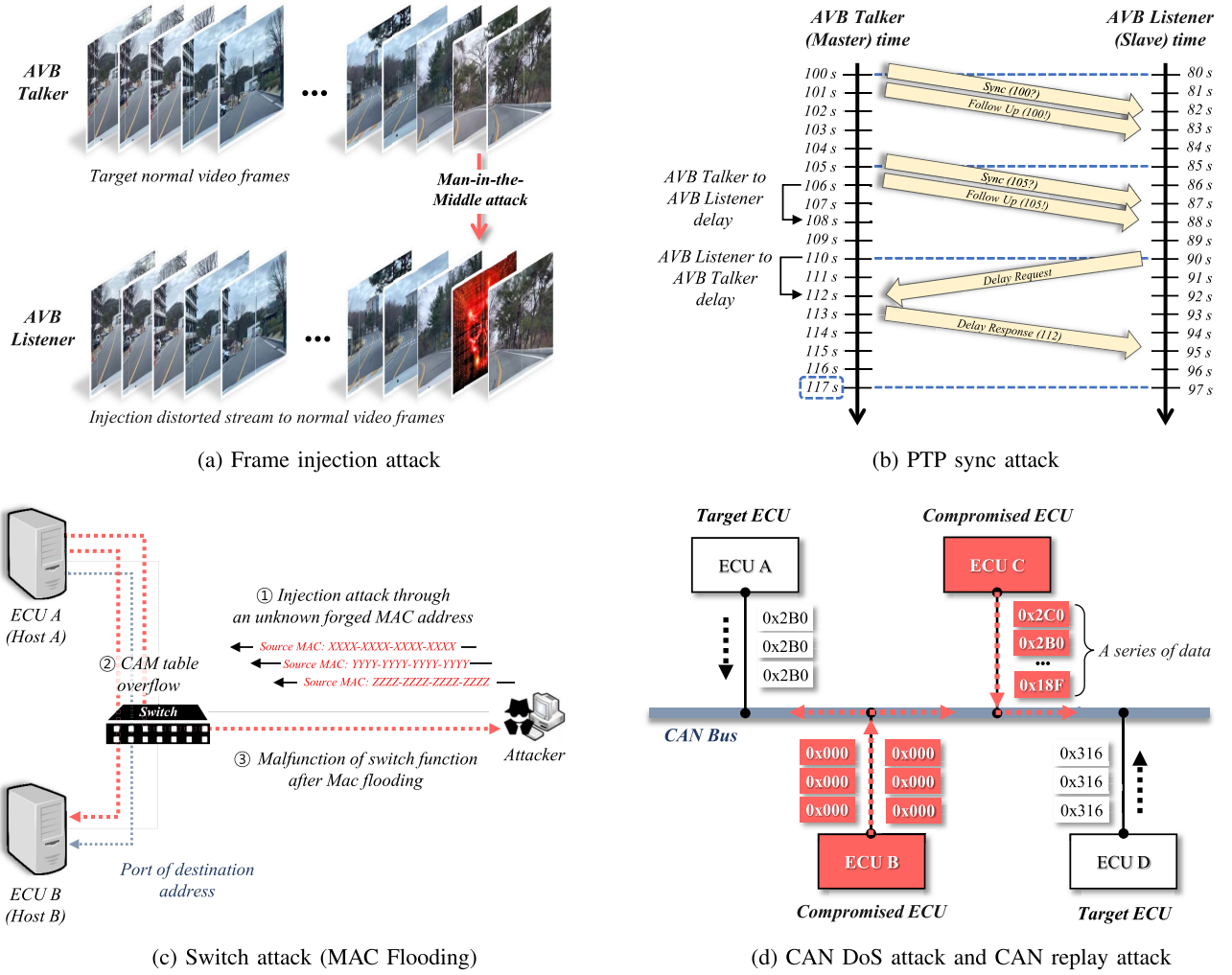


Fig. 2. Five types of attack scenarios.

overloaded (see Figure 2c). In this study, to reproduce the MAC flooding attack, we injected packets randomly composed of MAC and IP addresses for the source (src) and destination (dst) into the Automotive Ethernet network.

D. CAN DoS Attack and Can Replay Attack

As mentioned in Section II, considering its role as a multi-master of all ECU nodes connected to the CAN bus, the method of message broadcasting in a CAN bus topology and the priority principle for CAN messages constitute the main characteristics of the CAN protocol. However, these characteristics can be the basis for DoS, Replay, and Fuzzy types of attacks. First, the CAN DoS attack allows the highest priority arbitration ID transmitted from a malicious ECU node to occupy many of the resources allocated to the CAN bus based on the priority principle for CAN messages [30], [31]. Subsequently, in the CAN replay attack, the series of dumped data extracted from the normal traffic of the CAN bus are injected into the CAN bus again (see Figure 2d). It can attack the CAN bus system without conducting the reversing engineering for the functional information assigned to the arbitration ID. Moreover, the ID of injected CAN messages is equal to the arbitration ID value in the existing

normal CAN message frame. The CAN replay attack is performed by changing only the data field values in the CAN message frame. This makes it challenging to detect in a vehicle [32].

IV. METHODOLOGY

In this section, we explain the intrusion detection method with wavelet transform (WT) in the for Automotive Ethernet. The method comprises three processes of intrusion detection: 1) data extraction, 2) data preprocessing for the data reduction of the raw network traffic with the WT, and 3) intrusion detection and identification using customized ResNet deep learning algorithm. Figure 3 is an overview of the proposed intrusion detection method with WT. Depending on the data extraction process, the Automotive Ethernet traffic, including CAN, PTP, and AVTP is collected by the data extractor. In the data preprocessing step, the collected data is transformed into image data for application to the WT. The image data is input feed for two-dimensional discrete wavelet transform (2D DWT), which reduces the size of image data through 2D DWT. Consequently, the classification algorithms (such as the adjusted ResNet) apply the reduced data as input values.

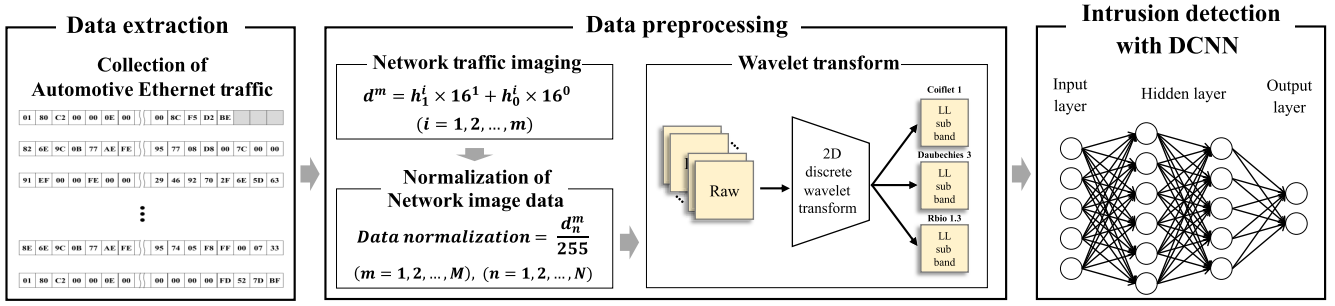


Fig. 3. Overview of the proposed intrusion detection system.

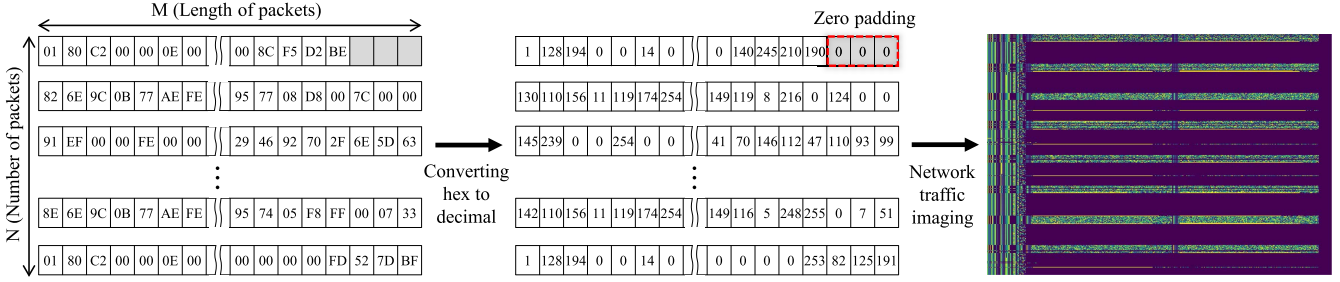


Fig. 4. Network traffic imaging.

A. Data Extraction

As a backbone network, Automotive Ethernet links the other networks (*i.e.*, CAN, LIN, FlexRay, and MOST) using a domain gateway. We focus on the CAN, AVB, and gPTP protocols in Automotive Ethernet. These protocols generate and transmit network traffic such as AVB stream data, gPTP sync, and encapsulated CAN messages. These various types of network traffic pass through the 100BASE-T1 switches to reach the destination in the end. We extracted the IVN traffic data using port mirroring with the 100BASE-T1 switch while all linked nodes communicate each. Moreover, to include the CAN message in Automotive Ethernet, we extracted the IVN traffic data by converting the CAN bus traffic to UDP packets.

B. Data Preprocessing

1) *Network Traffic Imaging*: The network packets have different sizes on the different protocols. For example, the packet size of AVTP is 434 bytes, while that of gPTP ranges from 60 to 90 bytes. The UDP packet that is converted to the CAN message size is 60 bytes. For this reason, the size of all packet types should be made same during the data preprocessing process of Automotive Ethernet. To satisfy this condition, we chose the number of network packets and the length of the network packets as $N \times M$, where N is the number of packets and M represents the length of packet (*i.e.*, 32×32 , 60×60 , 116×116 , 228×228 , and 452×452). If M is less than a fixed value, zero padding is added to match the height and width. On the contrary, if M is more than the fixed value, it takes only the number of packets corresponding to a fixed value and discards the remaining packets. Subsequently, the number of packets is added until the N value. In the next step, the packet data is converted

from hex to decimal. The hex value of the i -th byte is converted into the decimal value based on Eq. (1). For example, if the hexa value is '9B', the decimal value is calculated as: $d = 9 \times 16^1 + 11 \times 16^0 = 144 + 11 = 155$. Figure 4 shows the entire process of network traffic imaging set to the same height and width to provide a fixed input value to the classification algorithm.

$$d^m = h_1^i \times 16^1 + h_0^i \times 16^0, \text{ where } i \text{ is } 1, 2, \dots, m \quad (1)$$

2) *Normalization of the Network Image Data*: The decimal value from each byte in the network traffic ranges from 0 to 255. This continuous and wide range of input values can affect the feature value's saturation (no weight update) and weight convergence (fast gradient descent convergence) in the deep learning algorithm. Therefore, to mitigate the problem, we adjusted the scale of the byte value from zero to one using normalization. Through Eq. (2), d_n^m is an m -th value of decimal ($m = 1, 2, 3, \dots, M$) and n -th packets ($n = 1, 2, 3, \dots, N$).

$$\text{Network Data Normalization} = \frac{d_n^m}{255} \quad (2)$$

3) *Data Reduction Using Wavelet Transform*: Extensive input data makes a deep learning algorithm's parameter and model sizes massive. Moreover, it makes the deep learning model's training and testing speed more sluggish. As a type of data reduction method, WT ensures efficient utilization of limited resources and faster model learning. It is mainly used to analyze non-stationary data in signal processing and time-series data depending on the time-variant [33]. It has two types: continuous wavelet transform (CWT) and discrete wavelet transform (DWT). CWT has higher resolution and significant information about raw data as it analyzes the signal data continuously. Therefore, it requires more computation time to apply WT than DWT for signal analysis. Furthermore,

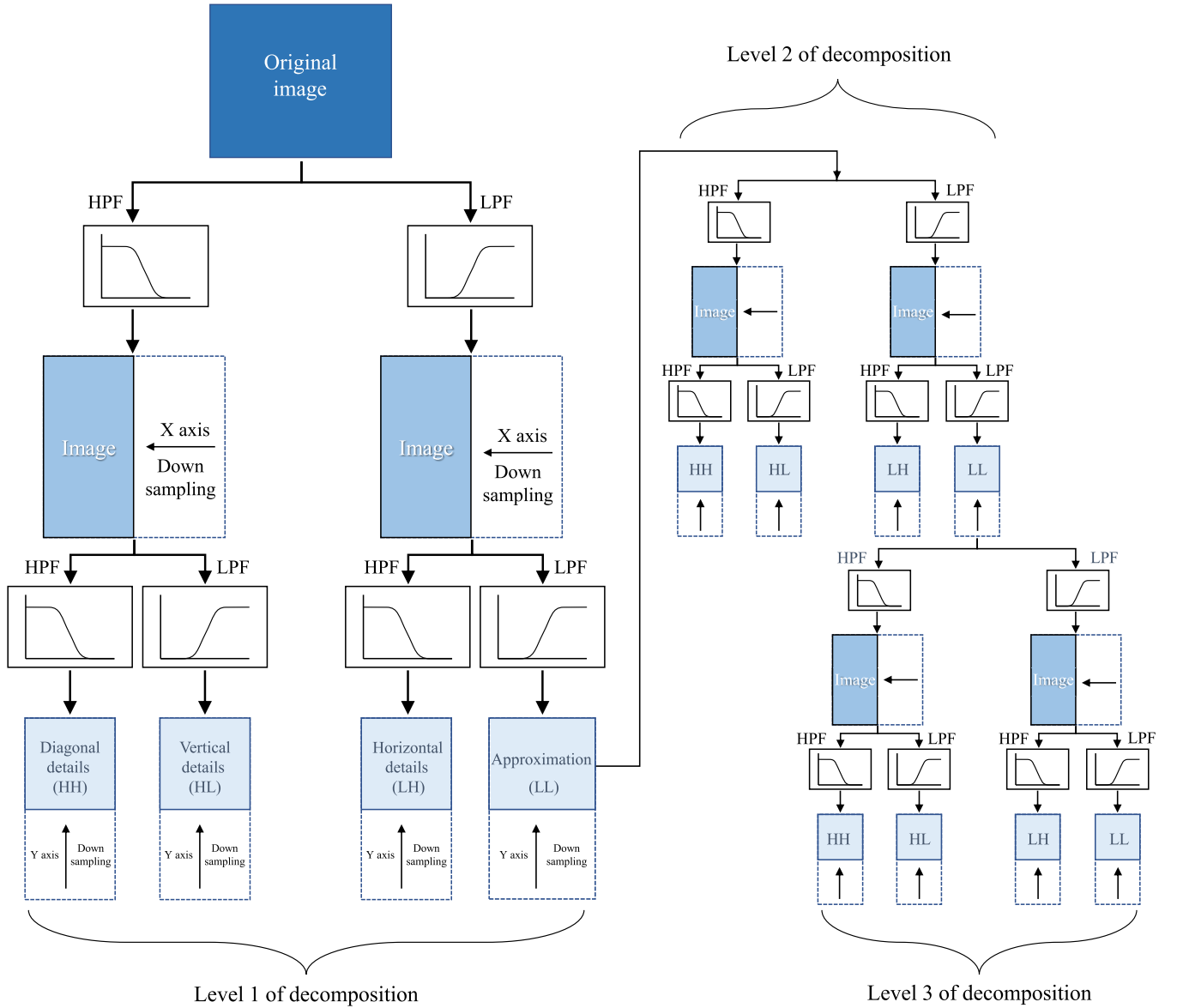


Fig. 5. Multi-resolution decomposition of wavelet transform.

CWT requires a large of resource space to store the algorithm's value. On the contrary, DWT incurs lesser time-cost and requires smaller resource space than CWT. Because of the different characteristics of the scale and shift factors of CWT and DWT, an appropriate comparison of suitability between both options is possible only under limited specific conditions. To numerically demonstrate the suitability of the DWT selection, we compared CWT to DWT experimentally depending on the same one-dimensional array. First, in the case of CWT, an average of 0.08350641 s is required when CWT is performed 1000 times for 512 numeric arrays after setting the scale factor in CWT. Whereas in the case of DWT, it takes 0.00016997 s. When data is compressed by performing WT through CWT, important information is included more than DWT. However, since it has many redundant factors, it requires more computation than DWT. Thus, DWT is more efficient than CWT in case of limited computational

resources [34], [35]. Based on the above strengths, we used the DWT (3) in our intrusion detection method.

As shown in Eq. (3), the original signal is $x(t)$, and n is an integer. Signal $x(t)$ can be decomposed by high-pass filter (HPF) and low-pass filter (LPF). Here, h_n and g_n depict the HPF and LPF, respectively. If signal $x(t)$ passes through HPF and LPF, the original signal is decomposed into detail ($cD1$, $y_{HPF}(k)$) coefficients and approximation ($cA1$, $y_{LPF}(k)$) as output. 2^j and $k2^j$ are the scaling and time-shifting parameters on the powers of two for the DWT, respectively.

$$DWT(j, k) = \frac{1}{\sqrt{|2^j|}} \int_{-\infty}^{\infty} x(t) \psi\left(\frac{t-k2^j}{2^j}\right) dt \quad (3)$$

Considering a possible 1D DWT of one-dimensional vector data, 2D DWT of two-dimensional vector data has the strength of information compression. Moreover, the coefficients from 2D DWT have the core information of the raw image data.

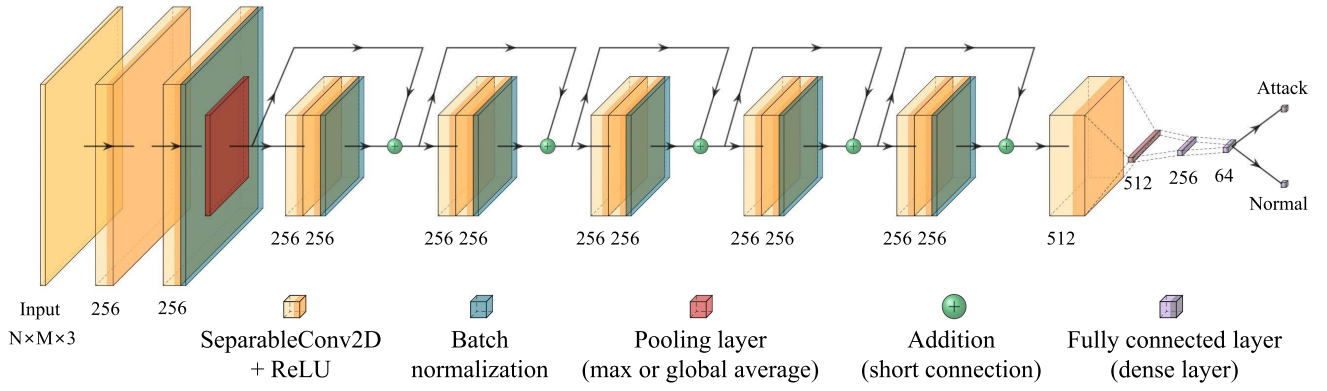


Fig. 6. Overview of deep convolutional neural network.

2D DWT comprises four sub-bands: low-low (LL), low-high (LH), high-low (HL), and high-high (HH). The LL sub-band has the core information about the original image, and the size is approximately 1/4 of the raw image size. Like the other coefficients, LH, HL, and HH are approximation and detail horizontal, vertical, and diagonal coefficients, respectively. In this study, to reduce the original dataset's computation complexity, we decomposed the original image data from the network traffic to the low-frequency sub-band (*i.e.*, LL) through the 2D DWT. Furthermore, we repeated the multi-resolution decomposition until the limitation of our image data input size was reached. Figure 5 depicts the multi-resolution decomposition of WT in detail. To feed into the input data from the original data's various representations, we used three wavelet filters (Coiflet 1, Daubechies 3, and Reverse biorthogonal 1.3) in 2D DWT [36]. As shown in Figure 5, the three LL data from the 2D DWT (being the possible RGB attributes of the image data) are layered. The three-layered LL data are then fed as input values into the deep learning algorithm (ResNet).

C. Intrusion Detection Using Convolutional Neural Network

DCNN is a robust algorithm that classifies an image with high accuracy. Moreover, ResNet, a deep learning algorithm, has shown high accuracy and low loss in various image classification challenges using residual weight value of each block of the algorithm [37]. In this study, we adopted ResNet as our main classification algorithm and adjusted its shape to reduce the number of ResNet parameters (see Figure 6).

1) *Convolutional Layer*: To recognize the image data, the convolutional layer extracts the features in conjunction with multiple convolution kernels for presenting higher-level features. The weight sharing from the convolution operation has three characteristics: the local pattern extraction, spatial learning hierarchies of feature patterns, and reduction in the number of parameters. For our neural networks, we used residual network and separable convolutional layer instead of the default convolutional layer to increase the efficiency by reducing the number of weight parameters. The ResNet's convolutional layer is defined as:

$$y = F(x, \{X_i\}) + x \quad (4)$$

where x and y are the input and output values of the convolutional layer, respectively. The function $F(x, \{X_i\})$ is a residual

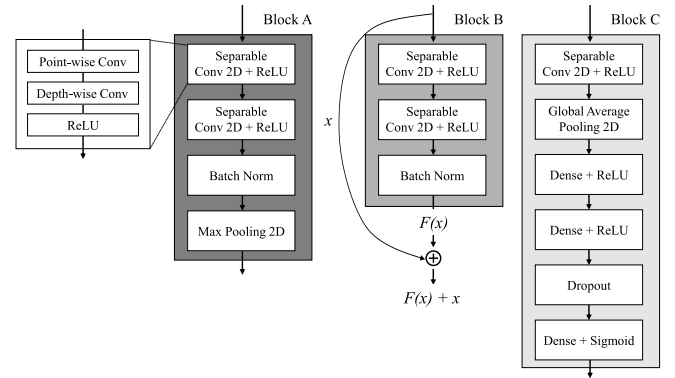


Fig. 7. Conv Blocks in TOW-IDS.

mapping, and the two feature maps, residual, and x perform element-wise addition channel by channel.

2) *Pooling Layer*: A pooling layer provides the downsampling of the feature maps' dimensionality. There are two most popular pooling operations: max and average pooling; both output the maximum and average values in the patch from the input feature maps, respectively. To map the input feature maps into a single vector, the global average pooling is used in Conv Block C. Max pooling and global average pooling are defined in (5) and (6), respectively.

$$s_j = \max_{i \in R_j} a_i \quad (5)$$

$$s_j = \frac{1}{|R_j|} \sum_{i \in R_j} a_i \quad (6)$$

where s_j represents the pooled feature maps at location j ; a_i is a set of feature map at location i ($a_1, a_2, \dots, a_{R_j}$) within the pooling region R_j .

3) *Fully Connected Layer*: The output feature maps of the convolutional should be transformed into a one-dimensional vector to map the classes. The flattened features from the fully connected layer (known as the dense layer) are connected to the probabilities for the classes in the classification. The dense layer is used in Conv Block C.

4) *Dropout and Batch Normalization*: In general, overfitting is a significant issue that decreases the model's

performance. Common methods that reduce overfitting are batch normalization and dropout. Batch normalization is an adaptive normalization method of the subsequent layers' input values with the scaling of the distribution of the input values [38]. Alternatively, dropout is used to select the activation function randomly in the training process [39]. Both ways make models less overfitted and more stable.

5) *Activation Function*: The vanishing gradient problem in the traditional neural network makes weight learning difficult because the increasing number of layers causes lower parameter value changes. ReLU, as an activation function, is useful for addressing this problem. It calculates the function 0 or max of the input value. In our neural networks, ReLU follows every convolutional and dense layers. We also used the sigmoid activation function to obtain the probabilities of the two classes from the final input weights of the dense layer. The output of the sigmoid activation function ranges from zero to one. The sigmoid function exists in the last layer following the dense layer. The ReLU and sigmoid activation functions are defined in (7) and (8), respectively.

$$f(x) = \max(0, x) \quad (7)$$

$$f(x) = (1 + e^{-x})^{-1} \quad (8)$$

The CNN algorithm's architecture mainly includes three Conv blocks: A, B, and C. Figure 7 shows Conv blocks A, B, and C in our proposed model. Conv block A is composed of two SeparableConv2D layers (256 channels, three strides, and ReLU activation function), batch normalization, and max-pooling layers. Conv block A's weight value is summed with the previous Conv block's output. To improve the testing speed (*i.e.*, intrusion detection), we did not utilize the default convolution layer; instead, we utilized a separable one that uses depth-wise and point-wise convolutions [40]. The separable convolution layer can continuously reduce the computation size using the depth-wise and point-wise convolutions. Five Conv block B are included in the deep neural network architecture. Conv block B comprises two SeparableConv2D layers (256 channels, three strides, and ReLU activation function). It consists of the batch normalization layer to mitigate the overfitting of the deep neural network. Conv block C is the final block in our proposed model. It has one SeparableConv2D layer (512 channels, three strides, and ReLU activation function), one global average pooling layer, two dense layers (256 and 64 channels with ReLU activation function), one dropout layer, and one dense layer (two channels and sigmoid activation function). As shown in algorithm 1, the proposed intrusion detection model is based on Normal and Abnormal datasets. The training dataset contains the labeled packets as Normal, Frame injection, PTP sync, Switch (MAC Flooding), CAN DoS, and CAN replay attacks, as shown in the attack scenarios in Section III.

V. EXPERIMENT AND EVALUATION

Section V-A briefly describes the equipment setup. In Section V-B, we present the experimental results. Section V-C compares the performance between those with and without the WT.

Algorithm 1 Intrusion Detection

Input: 'Packet'

Output: 'Attack' or 'Normal' log

```

1:  $M_{limit} \leftarrow$  Length of packets
2:  $N_{limit} \leftarrow$  Number of packets
3: Pkt_Bucket  $\leftarrow$  Packet buffer
4:  $Limit\_lv \leftarrow$  Multi-resolution level of wavelet transform
5:  $WT() \leftarrow$  Function of wavelet transform
6:  $mode \leftarrow$  'Train' or 'Test' mode selection
7:  $Trained\_Weight \leftarrow$  Trained model weight
  ▷ Data Generation
8: if  $len(Pkt\_Bucket) \leq M_{limit}$  then
9:   if  $len(Pkt\_Bucket) \geq M_{limit}$  then
10:     $Packet \leftarrow Pkt\_Bucket[:M_{limit}]$ 
11:   else
12:     $Packet \leftarrow ZeroPadding(len(Pkt\_Bucket), M_{limit})$ 
13:   end if
14:   Pkt_Bucket  $\leftarrow$  Packet      ▷ Packet length is  $M_{limit}$ 
15: else
16:   Pkt_image  $\leftarrow$  BytesToDecimal(Pkt_Bucket)
17:   Pkt_image  $\leftarrow$  Normalization(Pkt_image)
18:    $LL_1 \leftarrow WT(Pkt\_image, Limit\_lv, 'coiflet 1')$ 
19:    $LL_2 \leftarrow WT(Pkt\_image, Limit\_lv, 'DB 3')$ 
20:    $LL_3 \leftarrow WT(Pkt\_image, Limit\_lv, 'Rbio 1.3')$ 
21:    $3\_layered\_pkts \leftarrow Concatenate(LL_1, LL_2, LL_3)$ 
22: end if
  ▷ Applying a classification
23:  $IDS\_Model \leftarrow Model(Trained\_Weight)$ 
24: if  $mode == 'Train'$  then
25:    $IDS\_Model.fit(3\_layered\_pkts)$ 
26: else if  $mode == 'Test'$  then
27:    $Prediction \leftarrow IDS\_Model.predict(3\_layered\_pkts)$ 
28:   if  $Prediction == 'Attack'$  then
29:     $Log\_list \leftarrow 'Attack' \text{ log}$ 
30:   else
31:     $Log\_list \leftarrow 'Normal' \text{ log}$ 
32:   end if
33: end if

```

A. Setup for Dataset

As shown in Table I, to efficiently design and evaluate our proposed method, we extracted vehicle data of the Automotive Ethernet environment, including AVTP, gPTP, and UDP (UDP traffic are converted from CAN message). The collected data were divided into two datasets. One of the datasets contained Normal driving data without an attack. The other dataset included Abnormal driving data that occurred when an attack was performed. Table I shows the dataset size of Normal traffic and Abnormal traffic with five attack scenarios. The equipment setup used to extract vehicle data from the Automotive Ethernet environment was as follows. First, we simulated the experiment on machine with the following specs to assess the performance: 4790K CPU, 32GB RAM, and 2080 RTX GPU. Then, we used the Keras Python library for deep learning to apply the deep learning algorithm. Regarding parameter setting, we initialized 'adam' in the optimizer,

TABLE I
AUTOMOTIVE ETHERNET DATASET CONSTRUCTION
TO EVALUATE PERFORMANCE

Data type	Number of packets in dataset	
	Train dataset (ratio)	Test dataset (ratio)
Normal	9547,912 (83.5%)	660,777 (79.3%)
Frame injection	35,112 (2.1%)	16,962 (2.9%)
PTP sync	64,635 (3.3%)	26,013 (5.4%)
Switch	33,765 (2.1%)	16,809 (2.8%)
CAN DoS	85,466 (5.2%)	41,203 (7.1%)
CAN replay	29,847 (3.8%)	29,847 (2.5%)

binary cross-entropy in the loss function, and 100 epochs of the training iteration.

B. Experimental Results

The test dataset is composed of image data that are consecutive packets ($N \times M$). When one or more attack packets are included in the image data, we define it as an attack. To measure the performance, we set the metric true positive (TP) that is introduced as ‘True attack’ into the ‘Attack’ by our method. In contrast, if the proposed method predicted the ‘True normal’ as the ‘Attack,’ we regarded the metric as a false positive (FP). Similarly, if the method predicted the ‘True normal’ as ‘Normal,’ it was regarded as a true negative (TN). On the contrary, if our method predicted the ‘True attack’ as ‘Normal,’ we considered it as false negative (FN). The performance metrics are summarized as follows (see Eqs. (10)–(13)):

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN} \quad (9)$$

$$Recall = \frac{TP}{TP + FN} \quad (10)$$

$$Precision = \frac{TP}{TP + FP} \quad (11)$$

$$F - measure = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall} \quad (12)$$

$$AUC = \int_0^1 Pr[TP](v)dv, \\ \text{where } Pr[TP] \text{ is a function of } \\ v = Pr[FP] \quad (13)$$

To evaluate the performance of the proposed method, we measured the detection accuracy and the detection speed considering two aspects (*i.e.*, wavelet types and multi-resolution decomposition levels). Table II describes the accuracy metrics based on the wavelet type and multi-resolution decomposition level in detail. Accuracy metric results of TOW-IDS, whether WT preprocessing was performed or not, were examined according to image size. The result of reflecting a basic wavelet such as the ‘haar’ was added. The accuracy of our proposed method is 0.9232 in one level of the multi-resolution decomposition of the 32×32 image data, 0.9939 in two levels of multi-resolution decomposition of the 60×60 image data, 0.9787 in two levels of multi-resolution decomposition of the 116×116 image

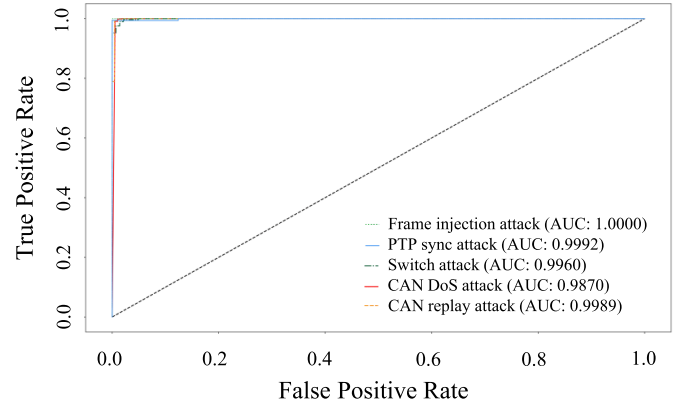


Fig. 8. ROC curves for TOW-IDS (Level 4 multi-resolution decomposition of 452×452 image data).

data, 0.9899 in one level of multi-resolution decomposition of the 228×228 image data, and 0.9965 in four levels of multi-resolution decomposition of the 452×452 image data. The best performance of the F-measure is 0.9974 in four levels of multi-resolution decomposition of the 452×452 image data using three-layered wavelets. Figure 8 shows the receiver operating characteristic (ROC) curve and area under curve (AUC) value to scrutinize the sensitivity and specificity. It focused on the four levels of multi-resolution decomposition of the 452×452 image data among the image sizes listed in Table II for five attack types. Table III describes the accuracy metrics between the proposed TOW-IDS and TOW-IDS with no WT based on multi-resolution decomposition level and image size.

The key to IVN intrusion detection is to consume few resources and incur low time-cost. We closely evaluated the performance of the proposed method, *i.e.*, the training and testing time-costs. Before measuring these time-costs, we measured the time-cost of preprocessing. The time-cost of preprocessing includes the network traffic imaging and normalization. The time-cost of one packet preprocessing is 0.00003827 s. For 452 packets, the time-cost of preprocessing is 0.01729804 s, and the time-cost of WT is 0.01164361 s. As a result, the total time-cost of 452 packet preprocessing is 0.02894165 s. Comparing these time-cost values, our proposed method spent approximately 0.001058 s to train an image data and approximately 0.00025 s to infer an image data. The ResNet algorithm without WT preprocessing (*i.e.*, raw image data) took approximately 0.353773 s to train an image data and approximately 0.039719 s to infer an image data. The time-cost of ResNet without WT was approximately 334 times for the time-cost of the proposed method in the four levels of multi-resolution decomposition of the 452×452 image data. Comparing our proposed method to the original dataset without WT preprocessing shows that the three-layered wavelet with WT is more efficacious in the training time (*i.e.*, low time-cost). The time-cost of WT is 0.011643610 s in the four levels of multi-resolution decomposition of the 452×452 image data, achieving the highest accuracy in our experiment. The time-cost measurement does not require a graphics card, and a high-speed CPU can draw better performance.

TABLE II
PERFORMANCE EVALUATION BY WAVELET TYPES AND MULTI-RESOLUTION DECOMPOSITION LEVELS. IN THE 1 WAVELET
(WITH 1 SUB-BAND AND 4 SUB-BANDS) CONDITION, WE CHOOSE THE 'HAAR' WAVELET

Wavelet type		3 wavelets with 1 sub-band		1 wavelet with 4 sub-bands		1 wavelet with 1 sub-band	
Image size		Accuracy	F-measure	Accuracy	F-measure	Accuracy	F-measure
32 × 32	Lv1	0.9232	0.9579	0.9199	0.9562	0.9178	0.9550
	Lv2	0.9939	0.9964	0.9769	0.9864	0.9872	0.9923
60 × 60	Lv1	0.9857	0.9917	0.9680	0.9808	0.9501	0.9704
	Lv2	0.9520	0.9726	0.9577	0.9759	0.9763	0.9858
116 × 116	Lv1	0.9787	0.9873	0.9879	0.9927	0.9799	0.9880
	Lv2	0.9760	0.9856	0.9689	0.9814	0.9653	0.9788
228 × 228	Lv1	0.9899	0.9937	0.9827	0.9894	0.9837	0.9903
	Lv2	0.9654	0.9790	0.9737	0.9837	0.9812	0.9886
452 × 452	Lv1	0.9812	0.9883	0.9825	0.9894	0.9791	0.9875
	Lv2	0.9833	0.9895	0.9708	0.9788	0.9631	0.9767
	Lv3	0.9718	0.9841	0.9781	0.9874	0.9841	0.9902
	Lv4	0.9934	0.9956	0.9894	0.9934	0.9944	0.9962
	Lv3	0.9959	0.9971	0.9936	0.9962	0.9909	0.9941
	Lv4	0.9965	0.9974	0.9921	0.9948	0.9908	0.9940
	Lv5	0.9861	0.9912	0.9859	0.9897	0.9590	0.9698

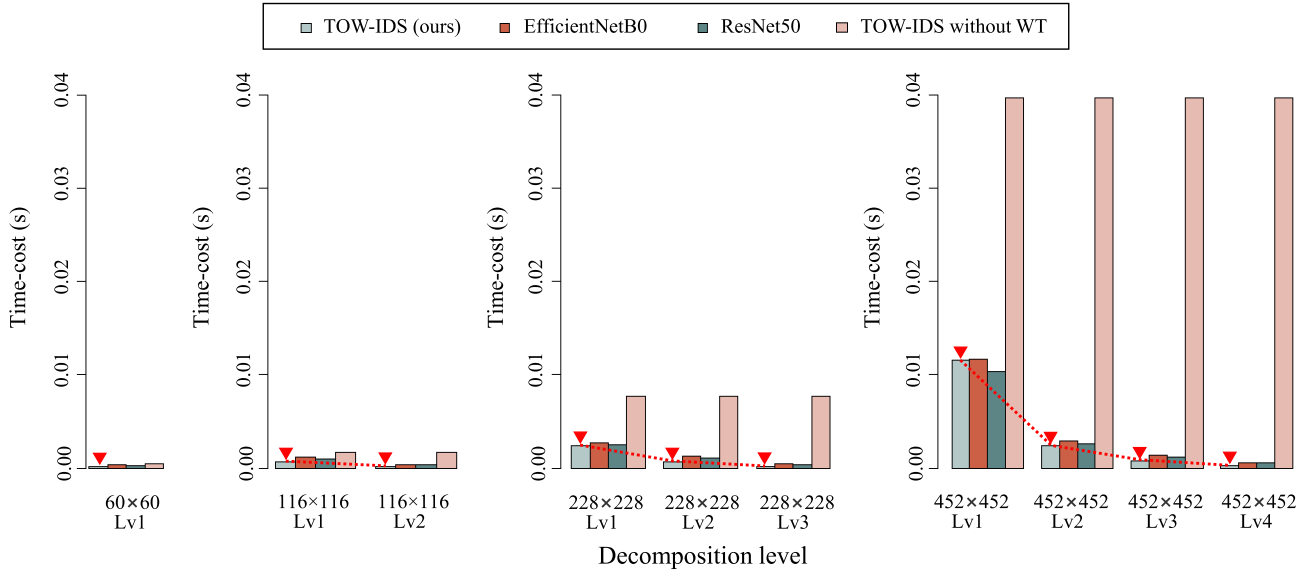


Fig. 9. Comparison of time-cost between the proposed method and other deep learning algorithms in testing perspective. Experiments with the proposed DCNN without WT were performed once in all the image sizes. The red dashed-line represents the decrease in time-cost based on the level of multi-resolution decomposition in WT.

C. Performance Comparison

To demonstrate the effectiveness of the proposed method, it was compared with the vanilla ResNet50 and EfficientNetB0 models in terms of time-cost incurred and the level of multi-resolution decomposition in WT. To find whether the application of WT affects the time-cost incurred, we added our customized DCNN for the raw dataset without WT (see Table IV). Figure 9 provides a comparison of time-cost between the proposed method and other deep learning algorithms from a testing perspective. Comparing the time-cost of the 452×452 network image data to the raw dataset and four levels of multi-resolution decomposed image dataset, the time-cost of the raw dataset is higher (approximately 167 times) than that of the four-level decomposed image dataset.

Furthermore, we measured Accuracy and F-measure to compare the performance between the proposed method and other deep learning algorithms (the vanilla ResNet50 and the vanilla EfficientNetB0). Then, we measured their model size and time-cost of the training and testing.

Table IV presents the results of the performance comparison. It can be seen from the table IV that the proposed method shows the best performance results, *i.e.*, 0.9965 accuracy and 0.9974 F-measure. It also incurs less time-cost than vanilla EfficientNetB0 (under 2.4 times) and ResNet50 (under 2.4 times). It was found that ResNet50 has lower performance results and lower time-cost efficiency compared to the proposed method. In the case of EfficientNetB0, although it has smaller parameter numbers, it shows lower performance results than other deep learning algorithms. The model size of

TABLE III
COMPARISON OF ACCURACY METRICS WITH AND WITHOUT THE WT PREPROCESSING

Methods	Image size	Decomposition level	
		Accuracy	F-measure
TOW-IDS (ours)	32 × 32	0.9232 (Lv1)	0.9579 (Lv1)
	60 × 60	0.9939 (Lv1)	0.9964 (Lv1)
	116 × 116	0.9787 (Lv2)	0.9873 (Lv2)
	228 × 228	0.9899 (Lv1)	0.9937 (Lv1)
	452 × 452	0.9965 (Lv4)	0.9974 (Lv4)
TOW-IDS with no WT	32 × 32	0.9244	0.9579
	60 × 60	0.9634	0.9585
	116 × 116	0.9786	0.9875
	228 × 228	0.9922	0.9953
	452 × 452	0.9720	0.9842

TABLE IV
COMPARISON OF ACCURACY METRICS PERFORMANCE DEPENDING ON THE MODEL SIZE

Methods	Accuracy	F-measure	Model size (MB)
TOW-IDS (ours)	0.9965	0.9974	11.3
ResNet50	0.9651	0.9785	270.3
EfficientNetB0	0.9603	0.9776	46.8
TOW-IDS with no WT	0.9720	0.9842	11.3

TOW-IDS is smaller than that of ResNet50 and the EfficientNet. The number of total parameters for TOW-IDS is 978,284. By comparison, the number of total parameters for ResNet50 and EfficientNetB0 are 23,589,761, and 4,050,845—far more than that for TOW-IDS. In summary, the efficient architecture of TOW-IDS shows high performance in terms of accuracy and time-cost, as well as a small model size and requires smaller number of parameters. Our proposed method has a much lower time-cost than TOW-IDS and DCNN without WT.

VI. CONCLUSION

Currently, studies on security related to Automotive Ethernet are relatively still in an early stage and based on the characteristics of the existing Ethernet. Further, they mainly focus on detection against known attack types depending on the protocol type. For this reason, future research for detecting unknown attacks and to identify their attack types in Automotive Ethernet is very meaningful and important. Particularly, studies on intrusion detection on heterogeneous networks of state-of-the-art technologies installed inside and outside a vehicle will be a significant issue that will need to be studied continuously in the future. This study proposed a method for detecting and identifying abnormalities in Automotive Ethernet based on wavelet transform and customized DCNN that shows high image classification performance. Above all, the method significantly reduces the time-cost of learning data to detect abnormalities in an Automotive Ethernet network. Furthermore, our framework focuses on the integrated data created between the heterogeneous networks, including AVTP, gPTP, and UDP, considering various sensing and communication devices that can be installed inside and outside a vehicle. The results of this study will contribute to the understanding of how to define new attack types against heterogeneous networks

and how to correctly process Automotive Ethernet data for vehicle security and driver safety.

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